Completing Claude's Cycles: Multi-Agent Structured Exploration on an Open Combinatorial Problem

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Abstract

We solve the Hamiltonian decomposition problem for $\mathbb{Z}_m^3$ Cayley digraphs for all $m > 2$, completing the even case left open by Knuth [1]. The odd case is resolved independently via a different construction than Knuth's, requiring 5 explorations with no human intervention compared to 31 explorations with significant human guidance. The even case yields a closed-form construction—$m-2$ bulk layers plus two structured repair layers—that is simpler than previously known computational solutions.

The constructions are produced by two LLM agents operating under a shared structured exploration prompt ("Residue"), with an orchestrator agent (also Claude Opus 4.6 Thinking) routing data and tools between them. The same prompt applied to two models—GPT-5.4 Thinking (Extra High) and Claude Opus 4.6 Thinking—produces radically different strategies: one agent reasons top-down and produces symbolic proofs; the other reasons bottom-up and builds solvers. Neither solves the even case alone. The combination does. The three exploration logs—two agent-level, one orchestrator-level—constitute a controlled case study in LLM mathematical reasoning, with implications for prompt design, multi-agent coordination, and the nature of the conception bottleneck in automated theorem proving.

Construction code and verification suite: `https://github.com/no-way-labs/residue`.

1 Introduction

Consider the Cayley digraph on $\mathbb{Z}_m^3$ with generators e_1, e_2, e_3, where e_r increments the r-th coordinate modulo m. Each vertex has out-degree 3 and in-degree 3. The *Hamiltonian decomposition problem* asks: for which $m > 2$ can the $3m^3$ arcs be partitioned into three directed Hamiltonian cycles of length m^3?

This problem arose while Knuth was writing about directed Hamiltonian cycles for a future volume of *The Art of Computer Programming* [6]. Knuth solved the case $m = 3$ and posed the general problem as an exercise. Filip Stappers discovered empirical solutions for $4 \le m \le 16$, making the existence of decompositions highly plausible. Stappers then posed the problem to Claude Opus 4.6, Anthropic's hybrid reasoning model. Over the course of 31 explorations with significant human coaching, Claude found a construction valid for all odd m. Knuth verified the construction, proved its correctness, and published the result as "Claude's Cycles" [1], identifying exactly 760 valid "Claude-like" decompositions for odd $m > 1$. The even case was left open.

In the postscript to the revised paper (March 4, further revised March 6), Knuth reported a rapid sequence of developments: Stappers put Claude Opus 4.6 to work on the even case, producing a search-based solution via ORTools CP-SAT [7]; Ho Boon Suan produced a computational construction for even $m \geq 8$ using GPT-5.3-codex [8], describing the pattern as "far more chaotic" than the odd case; Kim Morrison formalized Knuth's odd-case proof in Lean 4 [9]; and Exocija found what Knuth called "probably the simplest possible" odd-case construction by interacting with GPT 5.4 and Claude 4.6 Sonnet [10]. Ho subsequently used GPT-5.4 Pro to produce a complete 14-page proof of [8]'s correctness without human editing [11]. Finally, Knuth acknowledged the present work, describing the even-case construction as "considerably simpler" than [8] and noting that the process analysis has "potentially significant implications for how new problems can be tackled and resolved in the future."

This paper makes two contributions:

(i) **Mathematical.** We provide closed-form constructions for both parities: an independent odd-case construction (different from Knuth's, with a symbolic proof) and a new even-case construction valid for all even $m \geq 4$. The even construction has the form: $m-2$ bulk layers of a constant matching, one structured staircase layer, and one terminal row-block layer. Both constructions are verified computationally for all $m \leq 2{,}000$ of the appropriate parity, with additional round-map spot checks passing at $m = 5{,}000$, 10,000, 20,000, and 30,000.

(ii) **Methodological.** We document how these constructions were produced by two LLM agents operating under a shared structured exploration prompt, with an orchestrator agent (a third Claude Opus 4.6 Thinking instance) mediating between them. The same prompt applied to GPT-5.4 Thinking and Claude Opus 4.6 Thinking produced radically different mathematical strategies. The full exploration logs are included as appendices.

The dual contribution is deliberate. The mathematical result is a solved problem; the methodological result is a documented process. We believe the logs are as important as the theorems—they provide a controlled comparison of LLM mathematical reasoning under identical scaffolding conditions.

2 The Problem and Prior Work

2.1 Formal Statement

Definition 2.1. Let $\Gamma_m = \mathrm{Cay}(\mathbb{Z}_m^3, \{e_1, e_2, e_3\})$ be the Cayley digraph on $\mathbb{Z}_m^3$ with generators $e_r : (i, j, k) \mapsto (i + \delta_{r1}, j + \delta_{r2}, k + \delta_{r3}) \pmod m$. A *Hamiltonian decomposition* of Γ_m is a partition of its $3m^3$ arcs into three directed Hamiltonian cycles C_1, C_2, C_3, each of length m^3.

Equivalently, at each vertex $v \in \mathbb{Z}_m^3$, one assigns a permutation $\sigma(v) \in S_3$ determining which of the three outgoing arcs belongs to which cycle. The constraint is that each cycle C_r visits every vertex exactly once.

2.2 Fiber-Coordinate Reformulation

The key reformulation, discovered independently by both our agents and by Knuth's Claude, passes to *fiber coordinates*:

$$s = i + j + k \pmod m, \quad x = i, \quad y = j. \tag{1}$$

In these coordinates, a vertex is (s, x, y) with $k = (s - x - y) \bmod m$, and the three generators become layer transitions from s to $s + 1$:

$$X : (x, y) \mapsto (x + 1, y), \tag{2}$$
$$Y : (x, y) \mapsto (x, y + 1), \tag{3}$$
$$I : (x, y) \mapsto (x, y). \tag{4}$$

At each layer transition $s \to s + 1$, the three available moves are exactly X, Y, and I, and a Hamiltonian decomposition corresponds to choosing, at each (x, y, s), which of $\{X, Y, I\}$ is assigned to which cycle—an ordered 1-factorization of the bipartite graph on each layer whose vertex classes are the two copies of $\mathbb{Z}_m^2$ (at layers s and $s + 1$) and whose three perfect matchings are I, X, and Y.

2.3 The Round-Map Criterion

For each cycle r, let $M_s^{(r)}$ be the matching (a permutation of $\mathbb{Z}_m^2$) assigned to color r at layer s. The *round map* for color r is the composition

$$P_r = M_{m-1}^{(r)} \circ M_{m-2}^{(r)} \circ \cdots \circ M_0^{(r)}, \tag{5}$$

which maps (x, y) at layer 0 back to (x', y') at layer 0 after traversing all m layers. Cycle r is Hamiltonian if and only if P_r is a single m^2-cycle on $\mathbb{Z}_m^2$.

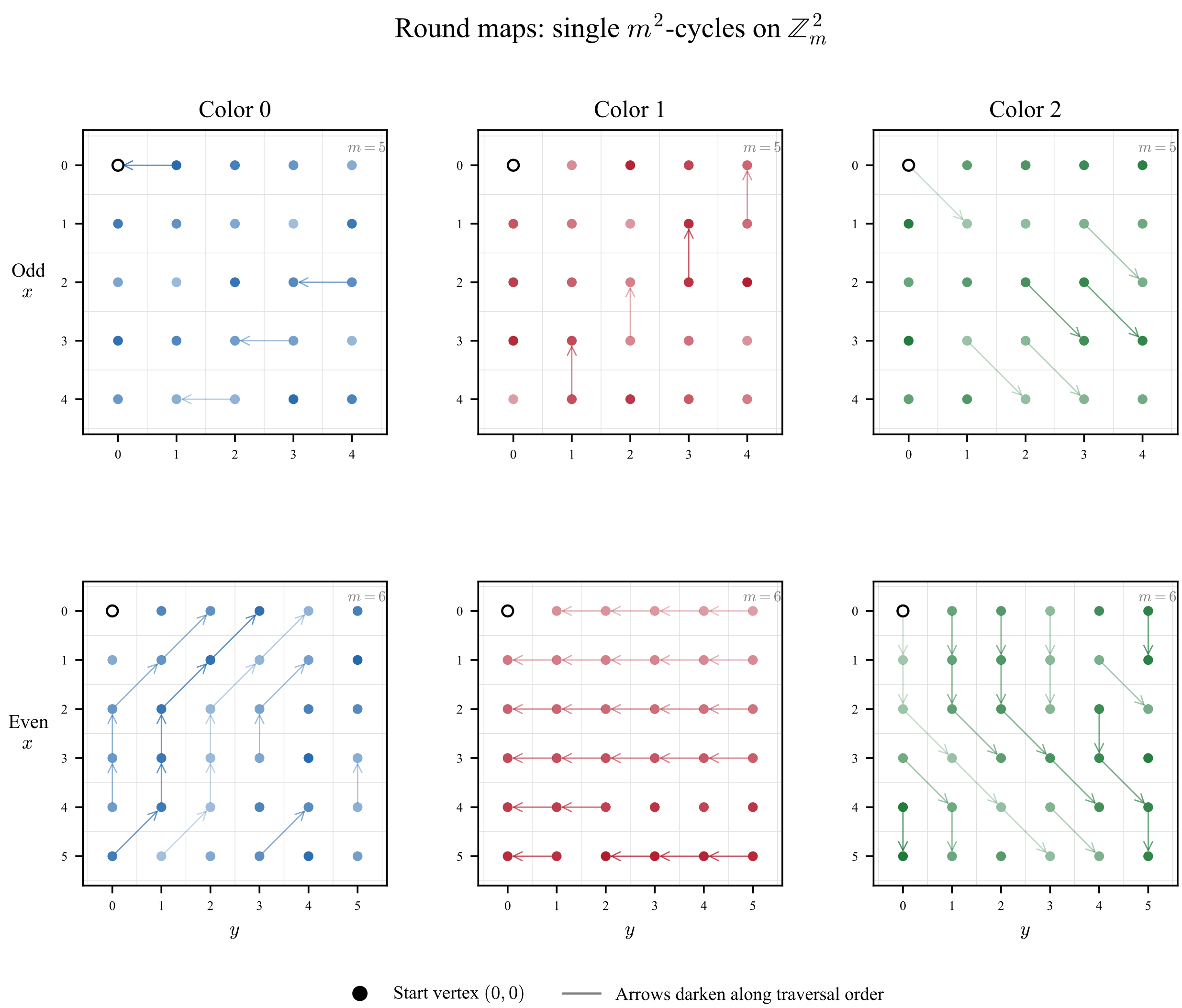

Figure 1: Round maps for the odd case ($m = 5$, top) and even case ($m = 6$, bottom). Each panel shows the single m^2-cycle traced by one color's round map on $\mathbb{Z}_m^2$. Arrows darken along the traversal order; the open circle marks the start vertex $(0, 0)$. All six round maps are verified to be single cycles.

2.4 The 2D Diagonal Gadget

Fix the diagonal $D = \{(x, y) : x + y \equiv 0 \pmod{m}\}$ in $\mathbb{Z}_m^2$. Define two permutations:

$$A : \text{use } X \text{ off } D, \text{ and } Y \text{ on } D, \tag{6}$$

$$B : \text{use } Y \text{ off } D, \text{ and } X \text{ on } D. \tag{7}$$

In rotated coordinates $u = x + y$, $v = x$:

$$A(u, v) = \begin{cases} (u + 1, v + 1) & \text{if } u \neq 0, \\ (u + 1, v) & \text{if } u = 0; \end{cases} \tag{8}$$

$$B(u, v) = \begin{cases} (u + 1, v) & \text{if } u \neq 0, \\ (u + 1, v + 1) & \text{if } u = 0. \end{cases} \tag{9}$$

Both A and B are Hamiltonian cycles on $\mathbb{Z}_m^2$ for every $m \geq 2$.

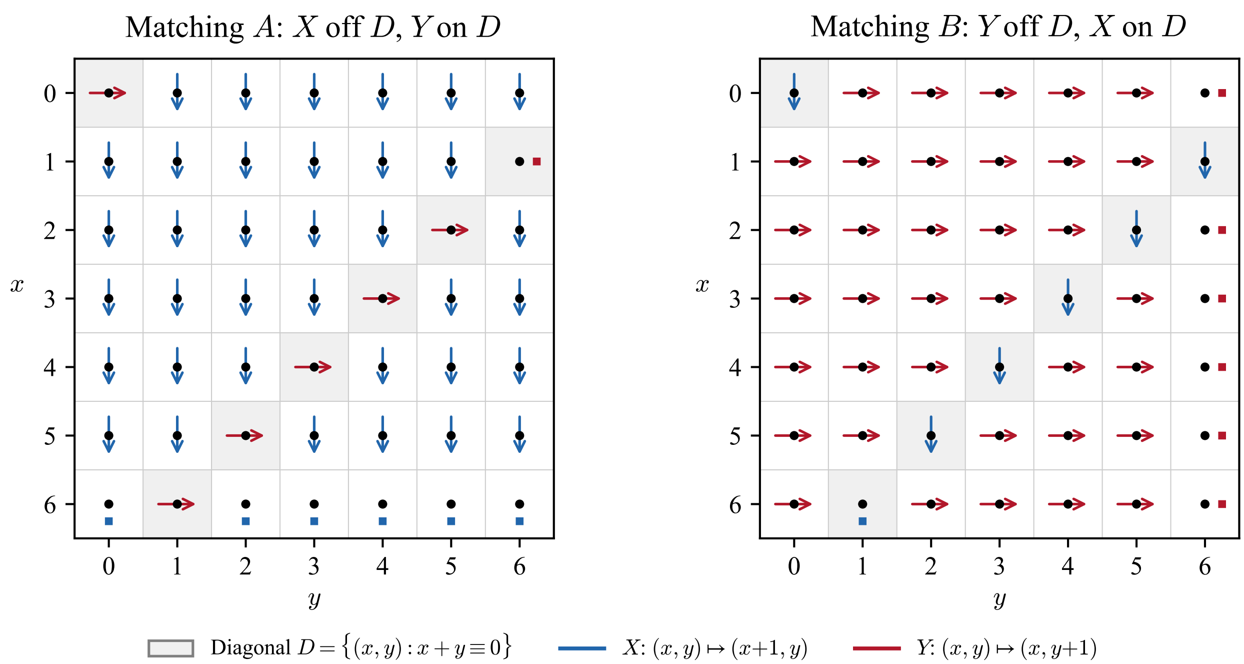

Figure 2: The diagonal gadget on $\mathbb{Z}_7^2$. Matching A uses X (blue, vertical) off the diagonal $D = \{(x, y) : x + y \equiv 0\}$ and Y (red, horizontal) on D. Matching B reverses the roles. Both are Hamiltonian cycles on $\mathbb{Z}_m^2$; the entire odd-case construction is built from words in A and B.

2.5 Prior Results

- **Aubert & Schneider (1982)** [2]: The undirected Cartesian product of cycles has a Hamiltonian decomposition. This does not cover the directed case.
- **Alspach's Conjecture (1984)** [3]: Every connected $2k$-regular Cayley graph on a finite abelian group has a Hamiltonian decomposition. Open in general.
- **Curran & Witte (1985)** [4]: $C_m \square C_m \square C_m$ has at least one Hamiltonian cycle. Does not give a full decomposition.
- **Darijani, Miraftab & Morris (2022)** [5]: Arc-disjoint Hamiltonian paths in products of directed cycles. Partial results; the full 3-cycle product decomposition was open.

- **Knuth (2026)** [1]: Odd-case construction via Claude Opus 4.6. Exactly 760 "Claude-like" decompositions valid for all odd $m > 1$. Even case left open. The March 6 revision acknowledges the present work's independent odd-case construction and even-case solution.

3 The Odd Case

We present an independent construction for all odd $m \geq 3$, different from Knuth's. Where Knuth's construction assigns a permutation based on the values of i, j, and $s = (i + j + k) \bmod m$ (distinguishing 0, $m - 1$, and "other"), ours works entirely at the layer level via the diagonal gadget of Section 2.4.

3.1 Layer Types

Define four layer types, each an ordered 1-factorization of $I + X + Y$:

$$P_0 = (I, B, A), \qquad P_1 = (B, I, A), \tag{10}$$
$$P_2 = (A, B, I), \qquad Q_2 = (B, A, I), \tag{11}$$

where the triple $(M^{(1)}, M^{(2)}, M^{(3)})$ specifies the matching assigned to each color at that layer.

3.2 The Construction

For odd $m \geq 5$, choose an odd residue s satisfying

$$\gcd(s, m) = 1 \quad \text{and} \quad \gcd(s + 2, m) = 1. \tag{12}$$

Such an s always exists: by the Chinese Remainder Theorem, take $s \equiv 2 \pmod 3$ on the 3-part of m, and $s \equiv 1 \pmod p$ for every other odd prime $p \mid m$. Set $t = (s - 1)/2$.

Use the following multiset of m layer types:

$$1 \times P_0, \quad 1 \times P_1, \quad t \times P_2, \quad (m - 2 - t) \times Q_2. \tag{13}$$

The ordering of layers does not affect the round-map analysis.

3.3 Hamilton Criterion

For each color, count the number of A-layers and B-layers seen:

$$C_1 : a_1 = t, \quad b_1 = m - 1 - t, \tag{14}$$
$$C_2 : a_2 = m - 2 - t, \quad b_2 = t + 1, \tag{15}$$
$$C_3 : a_3 = 2, \quad b_3 = 0. \tag{16}$$

Lemma 3.1 (Skew-Map Criterion). *For any word of length $n = a + b$ in the alphabet $\{A, B\}$, with a copies of A and b copies of B, the induced round map on $\mathbb{Z}_m^2$ has the form*

$$P(u, v) = \big(u + n,\ v + f(u)\big), \tag{17}$$

where $\sum_{u \in \mathbb{Z}_m} f(u) = b - a$. Consequently, $P^m(u, v) = (u, v + (b - a))$, and P is a single m^2-cycle if and only if both $\gcd(a + b, m) = 1$ and $\gcd(b - a, m) = 1$.

Proof. Each application of A or B increments u by 1 and increments v by either 1 (on the diagonal $u = 0$) or 0 (off the diagonal), with the choice depending on whether the letter is A or B. After $n = a + b$ applications, u has advanced by n, and v has accumulated shifts equal to the number of times the letter applied at the unique visit to $u = 0$ was B (contributing $+1$ if A) or A (contributing $+1$ if B). Summing f over all u yields $b - a$ because A contributes $+1$ at $u = 0$ and B contributes $+1$ at $u = 0$ (with opposite roles), and all other positions contribute identically.

After m full rounds, u returns to its original value and v has shifted by $m \cdot \sum f(u)/m = b - a$. The map P^m is therefore translation by $(0, b-a)$, which generates all of $\mathbb{Z}_m$ if and only if $\gcd(b-a, m) = 1$. The first return to the same u-value happens after $m/\gcd(n, m)$ applications of P, so the orbit has size m^2 if and only if $\gcd(n, m) = 1$ as well. □

Theorem 3.2 (Odd Case)**.** *For every odd $m \geq 5$, the layer schedule* (13) *yields a Hamiltonian decomposition of Γ_m. For $m = 3$, a separate explicit 3-layer decomposition exists.*

Proof. Apply Lemma 3.1 to each color:

$$C_1 : a_1 + b_1 = m - 1, \quad b_1 - a_1 = -(2t + 1) = -s; \tag{18}$$
$$C_2 : a_2 + b_2 = m - 1, \quad b_2 - a_2 = 2t + 3 = s + 2; \tag{19}$$
$$C_3 : a_3 + b_3 = 2, \quad b_3 - a_3 = -2. \tag{20}$$

Since m is odd, $\gcd(m - 1, m) = 1$ and $\gcd(2, m) = 1$. By the choice of s, $\gcd(s, m) = 1$ and $\gcd(s + 2, m) = 1$. All six coprimality conditions hold, so each round map is a single m^2-cycle.

The three round maps partition the arcs because at each layer, $(M^{(1)}, M^{(2)}, M^{(3)})$ is a permutation of $\{I, A, B\}$ (up to the identification of A and B with specific matchings), hence the three matchings cover $I + X + Y$ exactly. □

Remark 3.3. For the special case $m = 3$, where the general construction requires $m \geq 5$ (since $m - 2 - t \geq 0$ forces $m \geq 3$, but we also need the gcd conditions), an explicit 3-layer decomposition was found by exhaustive search:

$$\begin{aligned}
&\text{Layer } 0 : (I, X, Y), \\
&\text{Layer } 1 : \big(I|_{\{0,1\}} \cup Y|_{\{2\}},\ Y|_{\{0,1\}} \cup I|_{\{2\}},\ X\big), \\
&\text{Layer } 2 : \big(X,\ I|_{\{0\}} \cup Y|_{\{1,2\}},\ Y|_{\{0\}} \cup I|_{\{1,2\}}\big),
\end{aligned}$$

where subscripts indicate the rows on which each matching applies.

3.4 Verification

The construction was verified computationally for every odd m from 3 to 2001. The parameter choices observed were:

- $s = 1$, $t = 0$: used for all odd m not divisible by 3 (33 values);
- $s = 5$, $t = 2$: used for $m \in \{9, 27, 33, 39, 51, 57, 69, 81, 87, 93, 99\}$;
- $s = 11$, $t = 5$: used for $m \in \{15, 21, 45, 63, 75\}$.

3.5 Comparison with Knuth's Construction

Knuth's construction [1] is a vertex-level direction function: at each vertex (i, j, k), the rule depends on $s = (i + j + k) \bmod m$ and on whether i and j are 0, $m - 1$, or "other." Our construction works at the layer level: choose a sequence of layer types, and the vertex-level directions follow automatically from the diagonal gadget. The two approaches are genuinely different—they produce different decompositions for $m = 5$ and beyond—but they are morally related through the shared fiber-coordinate framework.

Knuth showed there are exactly 760 "Claude-like" decompositions (those depending only on whether i, j, s are 0, $m - 1$, or other). Our construction family is parametrized by the choice of s satisfying (12), giving a different (and typically smaller) family.

4 The Even Case

4.1 The Parity Obstruction

The odd construction fails for even m due to a fundamental parity obstruction.

Proposition 4.1 (Layer-Sign Parity Invariant)**.** *For even m, let $M_s^{(r)}$ be the layer-s matching for color r, and define the layer sign*

$$\tau_s = \prod_{r=1}^{3} \operatorname{sgn}\big(M_s^{(r)}\big). \tag{21}$$

Any Hamiltonian decomposition must satisfy

$$\prod_{s=0}^{m-1} \tau_s = -1. \tag{22}$$

In particular, the decomposition must contain an odd number of sign-negative layers.

Proof. The full color permutation P_r on m^3 vertices factors as $P_r(s, p) = (s + 1, M_s^{(r)}(p))$. Its sign is

$$\operatorname{sgn}(P_r) = (-1)^{m^2(m-1)} \prod_{s=0}^{m-1} \operatorname{sgn}\big(M_s^{(r)}\big). \tag{23}$$

When m is even, $(-1)^{m^2(m-1)} = 1$. A Hamiltonian cycle on m^3 vertices (where m^3 is even) has sign $(-1)^{m^3-1} = -1$. So each color requires $\prod_s \operatorname{sgn}(M_s^{(r)}) = -1$, and multiplying over all three colors gives $\prod_s \tau_s = (-1)^3 = -1$. □

In the same-diagonal layer family used for odd m, every layer uses only matchings from $\{I, A, B\}$, all of which have the same sign class. For even m, this forces τ_s to be constant across layers, making (22) impossible to satisfy with an even number of layers. This explains why the odd construction cannot extend to even m and why local search methods (Kempe-cycle swaps) that preserve layer signs get trapped.

4.2 Impossibility of $m = 2$

By exhaustive search (following Aubert & Schneider [2]), no Hamiltonian decomposition exists for $m = 2$.

4.3 The Even Construction

For every even $m = 2h \geq 4$, we provide an explicit fiber-coordinate construction. The small cases $m = 4, 6, 8$ are handled by exceptional tables found by computational search. For $m = 2h \geq 10$, the following uniform rule applies.

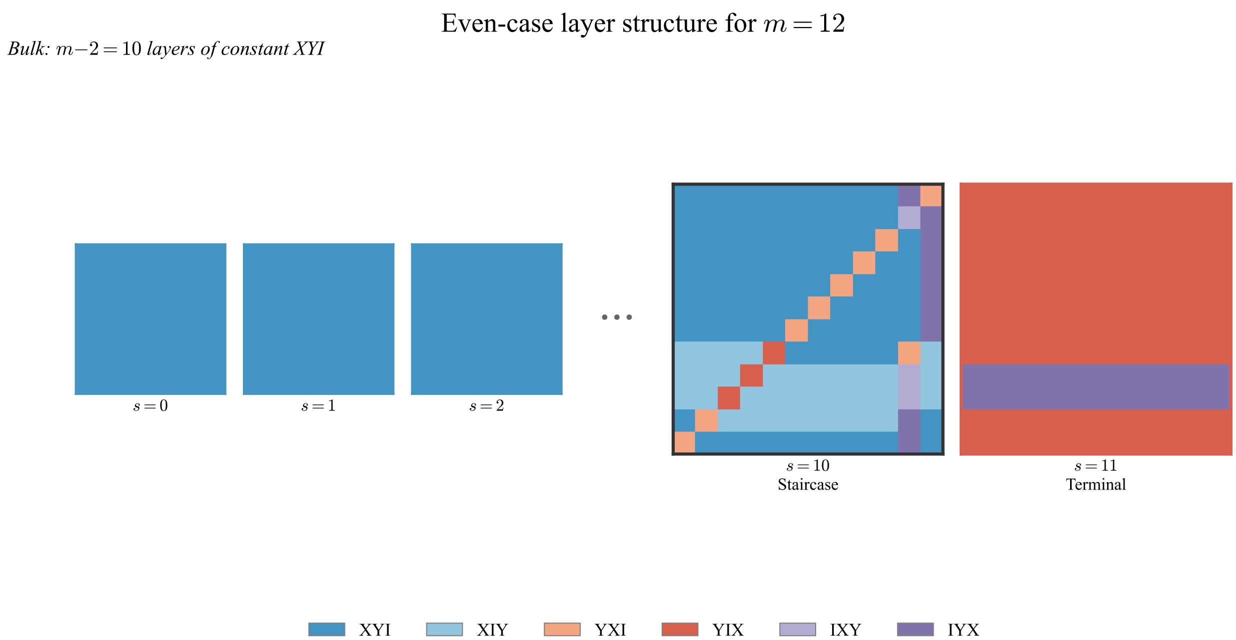

Figure 3: Layer structure of the even-case construction for $m = 12$. The first $m-2 = 10$ layers are constant XYI everywhere (blue). The penultimate staircase layer ($s = 10$) contains the diagonal walk and parity-dependent structure. The terminal layer ($s = 11$) is a row-block of YIX/IYX.

Definition 4.2 (Even Construction for $m = 2h \geq 10$)**.** The m layer transitions are specified as follows:

(a) **Bulk layers** ($s = 0, 1, \dots, m-3$): At every (x, y), use the constant assignment $(C_1, C_2, C_3) = (X, Y, I)$.

(b) **Terminal layer** ($s = m-1$): Use the constant row word YIX on every row except rows $h+2$ and $h+3$, where the row word is IYX.

(c) **Staircase layer** ($s = m-2$): This is the structured layer whose form depends on the parity of $h = m/2$.

Top half (rows $x = 0, 1, \dots, h$):

- Row 0: $\underbrace{XYI \cdots XYI}_{m-2}\ IYX\ YXI$.
- Row 1: $\underbrace{XYI \cdots XYI}_{m-2}\ IXY\ IYX$.
- Rows $2 \leq x \leq h$: $\underbrace{XYI \cdots XYI}_{m-1-x}\ YXI\ \underbrace{XYI \cdots XYI}_{x-1}\ IYX$.

Bottom half (rows $x = h+1, \dots, m-1$), parametrized by $d = m-1-x$ where $0 \le d \le h-2$:

If h is even:

- $d = h-2$: $\underbrace{XIY \cdots XIY}_{d}\ YIX\ \underbrace{XYI \cdots XYI}_{h-1}\ YXI\ XIY$.
- $d \in \{h-3, h-4\}$: $\underbrace{XIY \cdots XIY}_{d}\ YIX\ \underbrace{XIY \cdots XIY}_{m-d-3}\ IXY\ XIY$.
- $d = h-5$: $\underbrace{XYI \cdots XYI}_{d}\ YXI\ \underbrace{XIY \cdots XIY}_{m-d-3}\ IYX\ XYI$.
- $0 \le d \le h-6$: $\underbrace{XYI \cdots XYI}_{d}\ YXI\ \underbrace{XYI \cdots XYI}_{m-d-3}\ IYX\ XYI$.

If h is odd:

- $d = h-2$: $\underbrace{XIY \cdots XIY}_{d}\ YIX\ \underbrace{XYI \cdots XYI}_{h}\ IXY$.
- $d \in \{h-3, h-4\}$: $\underbrace{XIY \cdots XIY}_{d}\ YIX\ \underbrace{XIY \cdots XIY}_{m-d-2}\ IXY$.
- $d = h-5$: $\underbrace{XYI \cdots XYI}_{d}\ YXI\ \underbrace{XIY \cdots XIY}_{m-d-3}\ YIX\ XYI$.
- $0 \le d \le h-6$: $\underbrace{XYI \cdots XYI}_{d}\ YXI\ \underbrace{XYI \cdots XYI}_{m-d-3}\ IYX\ XYI$.

Example 4.3 (Staircase layer for $m = 10$)**.** For $m = 10$, we have $h = 5$ (odd). The 8 bulk layers ($s = 0, \dots, 7$) are constant XYI everywhere. The staircase layer ($s = 8$) is:

Top half (rows $x = 0, \dots, 5$):

Row	$y=0$	1	2	3	4	5	6	7	8	9
0	XYI	XYI	XYI	XYI	XYI	XYI	XYI	XYI	IYX	YXI
1	XYI	XYI	XYI	XYI	XYI	XYI	XYI	XYI	IXY	IYX
2	XYI	XYI	XYI	XYI	XYI	XYI	XYI	YXI	XYI	IYX
3	XYI	XYI	XYI	XYI	XYI	XYI	YXI	XYI	XYI	IYX
4	XYI	XYI	XYI	XYI	XYI	YXI	XYI	XYI	XYI	IYX
5	XYI	XYI	XYI	XYI	YXI	XYI	XYI	XYI	XYI	IYX

The "staircase" pattern is visible: YXI moves one position left with each row, while IYX anchors column 9.

Bottom half (rows $x = 6, \dots, 9$, parametrized by $d = 9 - x$):

Row	d	$y=0$	1	2	3	4	5	6	7	8	9
6	3	XIY	XIY	XIY	YIX	XYI	XYI	XYI	XYI	XYI	IXY
7	2	XIY	XIY	YIX	XIY	XIY	XIY	XIY	XIY	XIY	IXY
8	1	XIY	YIX	XIY	XIY	XIY	XIY	XIY	XIY	XIY	IXY
9	0	YXI	XIY	XIY	XIY	XIY	XIY	XIY	XIY	YIX	XYI

The terminal layer ($s = 9$) is constant YIX on all rows except rows 7 and 8, which use IYX.

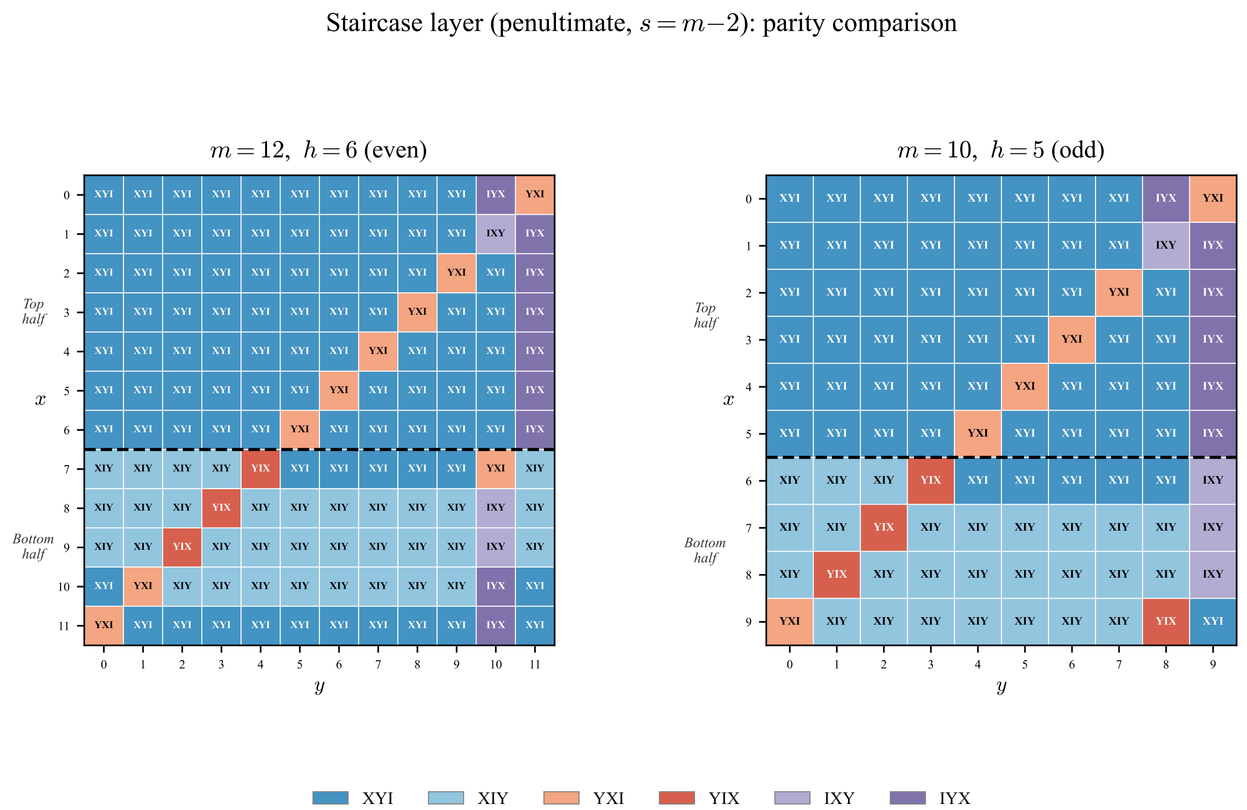

Figure 4: Staircase layer (penultimate, $s = m{-}2$) for $m = 12$ ($h = 6$, even) and $m = 10$ ($h = 5$, odd). Each cell is colored by its direction triple. The top half shows the diagonal walk (YXI descending one column per row, IYX anchoring the last column). The bottom half exhibits different patterns depending on the parity of h. The dashed line separates the two halves at row h.

Theorem 4.4 (Even Case)**.** *For every even $m \geq 4$, the construction of Definition 4.2 (using exceptional tables for $m = 4, 6, 8$ and the uniform rule for $m \geq 10$) yields a Hamiltonian decomposition of Γ_m.*

The proof of Theorem 4.4 is computational: for every even m from 4 to 2,000 (and spot-checked at $m =$ 5,000, 10,000, 20,000, and 30,000), the verification suite constructs all three round maps from the fiber-coordinate specification and confirms that each is a single m^2-cycle on $\mathbb{Z}_m^2$, that every arc is a valid Cayley digraph arc, and that the three cycles partition all $3m^3$ arcs. The construction code and verification suite are available at `https://github.com/no-way-labs/residue`. A symbolic proof of the even construction's correctness for all m remains open.

4.4 Why the Even Case Is Harder

Every approach to this problem—Knuth's Claude, Stappers, Ho Boon Suan, and both our agents—found the even case harder than the odd case. This is not an artifact of any particular method; it reflects genuine structural differences:

1. The parity obstruction (Proposition 4.1) means the odd-case diagonal gadget cannot be reused directly. Even constructions must break the single-coordinate regime.

2. The even construction requires *two* non-trivial layers (staircase and terminal) where the odd construction uses only one type of non-trivial layer in a count-based schedule.

3. The staircase layer exhibits a parity split (h even vs. h odd) with no analogue in the odd case.

4. The sign constraint means that local search methods (Kempe-cycle swaps) can be trapped in the wrong connected component of the solution space.

5 The Residue Prompt

The two agents operated under identical copies of a structured exploration prompt, which we call "Residue." The prompt prescribes documentation discipline but leaves mathematical strategy entirely open. Its design principles are:

1. **Structure the record-keeping, not the reasoning.** The prompt specifies *what to record* after each attempt (strategy, outcome, failure constraints, surviving structure, reformulations, concrete artifacts) but never suggests *what to try*.

2. **Make failures retrievable.** Each failed exploration produces a structured failure record: the specific structural reason it failed, the class of approaches it eliminates, and any partial results that survived. These records accumulate in a Strategy Register that prevents re-exploration of dead approaches.

3. **Force periodic synthesis.** Every 5 explorations, the agent must scan concrete artifacts across all previous explorations for patterns not yet commented on, check whether any reformulation suggests an untried approach, and write a brief synthesis. This is routine maintenance, not a signal of difficulty.

4. **Bound unproductive grinding.** If the Strategy Register has not changed in 5 explorations—no new eliminated classes, no new structural constraints, no new reformulations—the agent must state this explicitly and assess whether it is generating minor variations of already-eliminated approaches.

5. **Preserve session continuity.** At the start of each session, the agent re-reads the full log and Strategy Register before doing anything else. It does not start from scratch or re-derive established results.

The full prompt text is reproduced in Appendix A. Its relationship to Stappers's manual shepherding of Knuth's Claude is direct: the prompt automates the session-management role that Stappers performed by hand (reminding Claude to document progress, suggesting it review previous work, preventing aimless exploration).

6 Agent O's Trajectory

Agent O (OpenAI GPT-5.4 Thinking, set to Extra High) is a top-down symbolic reasoner. Given the Residue prompt and the problem statement, it proceeded as follows.

Key observations:

Table 1: Agent O exploration summary.

#	Outcome	Action
1	Key move	Fiber-coordinate reformulation. Recast the 3D problem as an m-layer skew product over $\mathbb{Z}_m^2$.
2	Success	Computed exact solutions for $m = 3$ and $m = 4$. Confirmed layer factorization as the right granularity.
3	Success	Symbolic diagonal family analysis. Found the Hamilton criterion: gcd conditions on (a_r, b_r) counts. Works for odd $m \geq 5$. Identified parity obstruction for even m.
4	Partial	Explicit odd construction completed. Even case: exhaustive search at $m = 4$ rules out single-transversal families.
5	Success	Verification sweep for odd m from 3 to 101. All pass. Odd case solved.

- **Five explorations, no human intervention.** The same milestone (odd case solved for all m) that Knuth's Claude reached in 31 explorations with significant coaching.[1]

- **Different construction.** Agent O's layer-count approach is algebraically cleaner than Knuth's vertex-level rule, but produces a different (smaller) family.

- **The Strategy Register prevented backtracking.** The fiber-coordinate reformulation was recorded at exploration 1 and elevated to a first-class structural tool via the Reformulations field. It was never revisited or re-derived.

- **The serpentine attractor was skipped entirely.** Knuth's Claude spent ∼10 explorations on serpentine constructions. Agent O never considered them—it went directly to the symbolic diagonal family.

6.1 Even-Case Push (Explorations 6–25)

After the odd case was solved, Agent O was directed to the even case with a 15-exploration budget. Key milestones:

- **Exploration 7:** Discovered the layer-sign parity invariant (Proposition 4.1). Found $m = 6$ by whole-layer search. This is a genuine structural contribution not in Knuth's paper.

- **Exploration 8:** Found $m = 8$ by the same method.

- **Exploration 9:** Stalled at $m = 10$. Best profile $[97, 3], [100], [100]$—a rigid basin, not a compute issue.

[1] Exploration count is an imperfect metric. Many of Claude's 31 explorations involved reasonable strategies—testing linear and quadratic functions, trying serpentine constructions—that happen to fail on this particular problem but might have succeeded on a different one. As Knuth notes [12], optimizing for any particular metric is often misleading. The comparison is reported as a factual observation, not a claim that fewer explorations is unconditionally better.

- **Exploration 13:** Identified the permutation-sign obstruction in the $m = 10$ near-hit: the defective color has sign $+1$ but needs -1.

- **Exploration 16:** Received fiber-coordinate tables from Agent C (see Section 8). Immediate pattern recognition: $m - 2$ bulk XYI layers plus two structured repair layers.

- **Exploration 21–22:** Used the seeded solver (adapted from Agent C's MRV solver) to obtain exact solutions for $m = 14$ and $m = 16$. Refitted the staircase rule.

- **Exploration 23:** Predicted $m = 18$ table from the fitted rule. Verified directly—all three round maps are single 324-cycles.

- **Exploration 25:** Verified the complete even construction for all even m from 4 to 100.

7 Agent C's Trajectory

Agent C (Anthropic Claude Opus 4.6 Thinking) is a bottom-up computational solver. Given the identical Residue prompt and problem statement, it followed a markedly different trajectory.

Table 2: Agent C exploration summary.

#	Outcome	Action
1–2	Success	Computational search for $m = 3$. Found solutions, identified cyclic symmetry.
3	Failure	d_0-table approach for $m = 4$: exhaustive search of all 3^{12} tables found 0 valid solutions.
4	Partial	Block structure analysis from $m = 3$ pattern. Does not scale.
5–9	Failure	Serpentine constructions. Extended dead end (5 explorations, vs. $\sim$10 for Knuth's Claude). Structured logging forced clear articulation of why serpentines fail.
10	Failure	Binary search with fixed serpentine cycle. Same fundamental incompatibility.
11–12	Partial	BFS permutation search finds $m = 4, 5$ but fails at $m = 6$ (exponential blowup).
13	Partial	Quotient approach via $\langle(1, 1, 1)\rangle$. Works for $m = 3$ only.
14	**Breakthrough**	MRV + forward checking. 67,000× speedup over BFS. Solutions for $m = 3$ through $m = 12$.
15	—	Literature search found Knuth's paper. Stopped.

Key observations:

- **The serpentine attractor was compressed.** Five explorations (7–9 plus variants), compared to $\sim$10 for Knuth's Claude. The Strategy Register's "What This Rules Out" field forced the agent to generalize each failure to a class, preventing minor-variation grinding.

- **The breakthrough was algorithmic, not mathematical.** MRV + forward checking is a standard CSP technique, but applying it to this specific search space—with dynamic variable ordering and domain pruning—was a creative engineering act that doesn't fit neatly into "conception" vs. "search."

- **More even-case data than any other single agent.** Solutions for $m = 4, 6, 8, 10, 12$—the raw material that later unblocked Agent O's even-case push.
- **Never found the fiber-coordinate framework.** Agent C noted the fiber/coset structure (Table 2, exploration 7) but did not build on it. The Reformulations field can preserve an insight but cannot force a model to recognize which insight is load-bearing.

8 The Orchestration

The orchestrator agent (a third Claude Opus 4.6 Thinking instance, directed by the author) made five key decisions, each documented in the meta-observation log (Appendix B).

8.1 Decision 1: Two Agents, Same Prompt

Launch two agents from different model providers on the identical problem with the identical prompt. No hints about odd/even split, no mathematical guidance. Benchmark: does the Residue prompt improve on the 31-exploration trajectory from Knuth's paper?

Outcome: Both agents solved the odd case (one in 5 explorations, one partially via computed examples). The prompt structured the record-keeping identically while the models diverged in reasoning style.

8.2 Decision 2: Directing Agent O to Even

After Agent O solved the odd case, direct it to the even case with a 15-exploration budget. Agent C was still running independently.

Outcome: Agent O discovered the layer-sign parity invariant (a new structural result) and found $m = 6, 8$, but stalled at $m = 10$.

8.3 Decision 3: Cross-Agent Data Transfer

Agent C had solutions for $m = 10$ and $m = 12$. Agent O had the structural framework but could not reach $m = 10$. The orchestrator asked Agent C to export its solutions in fiber-coordinate format (Agent O's native representation) and passed the resulting tables to Agent O.

Outcome: Immediate pattern recognition. Agent O identified the $m - 2$ bulk layers plus two repair layers within minutes of receiving the data. This was the critical move that neither agent could have made alone.

8.4 Decision 4: Tool Transfer

Agent O needed to verify predictions at $m = 14, 16$ but could not compute exact solutions. Agent C had the MRV solver. The orchestrator extracted the solver and placed it in Agent O's working directory.

Outcome: Agent O did not use the solver as given—it adapted it into a *seeded* solver, using its own structural predictions to constrain the search domain. The seeded solver is faster than either the raw MRV or Agent O's analytical approach alone.

8.5 Decision 5: Context About External Results

The orchestrator informed Agent O that other researchers (Stappers with Claude, Ho Boon Suan with GPT-5.3-codex) had found working even-case constructions but described them as "chaotic" and "far more complex" than the odd case. No constructions were provided.

Outcome: This provided motivation and calibration. Agent O's construction turned out to be simpler than both external solutions.

8.6 What the Orchestrator Did That the Prompt Could Not

1. **Recognized complementary strengths.** Agent O had theory; Agent C had data. Neither knew the other existed.
2. **Chose the transfer representation.** Fiber coordinates were Agent O's native language. Converting Agent C's output to fiber tables made the transfer immediate rather than requiring Agent O to parse a foreign format.
3. **Timed the transfers.** Data was sent after Agent O had stalled (not before, which would have been noise), and the tool was sent after Agent O had demonstrated it could use the data but lacked verification capacity.

9 Analysis

9.1 About the Prompt

(a) **Compression:** 5 explorations vs. 31 for the same milestone (odd case solved), though exploration count is a crude proxy (see footnote in Section 6). The structured logging forced each exploration to be maximally informative.

(b) **Bounded dead ends:** Agent C spent 5 explorations on serpentines (vs. ~10 for Knuth's Claude). The "What This Rules Out" field generalized failures to approach classes, preventing minor-variation grinding.

(c) **Graceful escalation:** Agent O's $m = 10$ stall produced a useful handoff specification, not a degradation. The structured failure record (specific basin, specific diagnosis, specific gap) made the cross-agent transfer actionable.

(d) **Model-independence of process, model-dependence of strategy:** Same scaffolding, radically different mathematical approaches. The prompt structures process, not thought.

9.2 About Multi-Agent Coordination

(a) **Complementary failure modes.** Top-down reasoning stalls where bottom-up computation grinds through; bottom-up computation misses structure that top-down reasoning sees. The agents failed in *different* ways on the same problem.

(b) **Data transfer as the key orchestration primitive.** Not instructions, not strategies, but *artifacts in a shared representation.* The fiber-coordinate tables were the bridge, and their format was chosen to match the recipient's framework.

(c) **Tool transfer as amplification.** The seeded solver was better than either the raw MRV or the analytical approach alone. Agent O improved Agent C's tool by combining it with its own structural knowledge.

(d) **The orchestrator role as irreducible (for now).** The decisions about when to transfer, what to transfer, and in what format required judgment that neither solving agent had. The orchestrator was itself an LLM (Claude Opus 4.6 Thinking), but directed by a human. Whether a single agent can play all three roles simultaneously remains an open problem.

9.3 About the Conception Bottleneck

Single-agent frameworks for LLM mathematical reasoning implicitly frame the problem as one agent overcoming a conception barrier. Our experiment complicates this picture:

(a) Agent O's odd-case solution was an *interpolation*: it recombined known techniques (fiber coordinates, skew maps, gcd criteria) in a novel arrangement. This is creative but within the space of existing mathematical tools.

(b) Agent C's MRV breakthrough was an *algorithmic* creative act: designing a better search algorithm. This doesn't fit neatly into "conception" vs. "search."

(c) The even case required *cross-agent synthesis*: structure from one model, data from another, integration by an orchestrator. This is not the generate-and-check loop of single-agent theorem proving. It is something more collaborative.

9.4 About the Problem Itself

(a) The even case is inherently messier than the odd case. Every approach—human, single-agent, multi-agent—found this. The parity obstruction is real, not an artifact.

(b) The layer-sign parity invariant (Proposition 4.1) is a genuine structural discovery with potential applications to related Cayley graph decomposition problems.

(c) The even-case staircase construction, while verified to $m = 2{,}000$ (and spot-checked to $m = 30{,}000$), lacks a symbolic proof. Whether this can be proved by the same agents (or any current AI system) is an open question.

(d) Two natural symmetries generate additional decompositions from any given one. First, one may *conjugate* the layer indexing: reversing the roles of layers 0 and $m-1$ (so that the terminal layer sits at $s = 0$ and the staircase at $s = m-1$, or any cyclic shift) yields a valid decomposition whenever the original is, since the round-map criterion depends only on the cyclic product of all m layers. Second, *reversing* a decomposition—traversing each Hamiltonian cycle backwards and reading directions after each step rather than before—gives another valid decomposition. These symmetries, noted by Knuth [12], enlarge the family of known decompositions but do not change the underlying construction.

10 Limitations and Open Questions

1. **The even-case proof is computational.** The construction of Definition 4.2 is verified for all even $m \leq 2{,}000$ (with spot checks to $m = 30{,}000$), but a symbolic proof of correctness for

all even m remains open. The staircase layer's parity-sensitive case structure makes direct symbolic analysis substantially harder than the odd case.

2. **The experiment has $n = 2$ agents.** Broader conclusions about prompt design and multi-agent coordination require more replications across different problems and model providers.

3. **The orchestration was reconstructed.** The orchestrator agent's decisions were documented in a meta-observation log written during and after the sessions, not in strict real-time. Future experiments should log orchestration decisions as they happen.

4. **Partial prior knowledge.** Knuth's paper existed before the experiment began, and the odd case was already solved. The experiment tested whether the Residue prompt could compress the path to the same result and extend it to the open even case. A fully open problem (where no partial solution is known to anyone) would be a harder test.

5. **Generalizability beyond mathematics.** Whether the multi-agent coordination pattern (complementary reasoning styles, structured data transfer, tool transfer with adaptation) generalizes to software engineering, scientific modeling, or other domains is untested.

6. **No formal verification.** Neither the odd nor even construction has been formalized in a proof assistant. The odd case appears tractable for Lean 4 formalization (the core skew-map criterion reduces to coprimality conditions in $\mathbb{Z}_m$). The even case, with its parity-sensitive staircase casework, would be substantially harder. Formal verification is planned for future work.

11 Conclusion

The mathematical result. The Hamiltonian decomposition of $\Gamma_m = \mathrm{Cay}(\mathbb{Z}_m^3, \{e_1, e_2, e_3\})$ exists for all $m > 2$:

- For $m = 2$: impossible (exhaustive search).
- For odd $m \geq 3$: closed-form construction via the diagonal layer schedule (Theorem 3.2), with symbolic proof of correctness.
- For even $m \geq 4$: closed-form construction via bulk layers plus staircase and terminal layers (Theorem 4.4), verified computationally for all $m \leq 2{,}000$, with spot checks to $m = 30{,}000$.

The methodological result. A structured exploration prompt (Residue) reduces the path to the odd-case milestone from 31 explorations to 5 (though exploration count is an imperfect metric—see Section 6), while enabling graceful failure documentation and cross-agent data transfer. Two agents with different reasoning styles—one top-down symbolic, one bottom-up computational—produce complementary artifacts that, when combined by an orchestrator agent (also Claude Opus 4.6 Thinking), solve a problem neither could solve alone.

The deeper point. The even-case solution required three participants with different capabilities: one that could see structure, one that could compute examples, and one that could recognize when the first needed what the second had produced. The prompt made each agent's output legible to the others. The orchestrator—itself an LLM, directed by the author—made the connections. All three participants were LLM agents. The result belongs to the system, not to any single component.

A The Residue Prompt

The full text of the structured exploration prompt used by both agents is reproduced below. Both agents received identical copies.

> **System Prompt: Long-Horizon Mathematical Investigation**
>
> You are working on a mathematical problem that may require many attempts across multiple sessions. Your job is to solve the problem. How you think about it is up to you. But how you *record* your work is not—follow the logging protocol below exactly.
>
> **The Exploration Log.** You maintain a file called `exploration_log.md`. After every substantive attempt—successful, failed, or abandoned—you update this file **before doing anything else**. No exceptions. Do not start a new attempt until the previous one is logged.
>
> There are two logging formats. Use the **full format** for substantive attempts. Use the **short format** for quick probes.
>
> *Short format:* Exploration number (probe), Strategy (one sentence), Outcome (SUCCEEDED/-FAILED/ABANDONED), Concrete Artifacts.
>
> *Full format:* Exploration number, Strategy, Outcome, Failure Constraint, What This Rules Out, Surviving Structure, Reformulations, Concrete Artifacts, Key Parameters, Open Questions.
>
> **The Strategy Register.** A running summary at the top of the log with three lists: Eliminated approach classes, Active structural constraints, Known reformulations. Updated after every exploration.
>
> **Session Continuity.** At the start of each session: read the log in full, read the Strategy Register, state your plan grounded in what you've learned. Do not start from scratch.
>
> **Periodic Synthesis.** Every 5 explorations: scan artifacts for unnoticed patterns, check reformulations for untried approaches, write a synthesis.
>
> **When You're Stuck.** After 3 consecutive failures with no new approach class: re-read all Concrete Artifacts, look for cross-strategy patterns, re-read Surviving Structure and Reformulations, write a synthesis. The solution may be in the residue of your previous failures.
>
> **When to Escalate.** If the Strategy Register hasn't changed in 5 explorations: assess whether you're generating minor variations, whether synthesis is producing new observations, and whether you can articulate the kind of insight you're missing. A well-documented dead end is more useful than an undocumented spiral.

B Meta-Observation Log

The orchestrator's meta-observation log is summarized in Section 8. The full log, including timestamped decisions and inter-agent data transfers, is available in the supplementary materials.

C Agent Exploration Logs

The complete exploration logs for both agents are available in the supplementary materials:

- **Agent O:** 28 explorations (5 for odd case, 23 for even case including Lean formalization attempts). Total: approximately 65,000 words.
- **Agent C:** 15 explorations. Total: approximately 25,000 words.

D Exceptional Even Tables

The fiber-coordinate tables for $m = 4, 6, 8$ are available in the supplementary materials. Each table specifies, for every layer s and every position (x, y), the triple $(C_1, C_2, C_3) \in \{I, X, Y\}^3$ defining the decomposition.

E Verification Code

Python verification code for both the odd and even constructions is available at `https://github.com/no-way-labs/residue`. The verifier constructs the three round maps from the fiber-coordinate specification and checks that each is a single m^2-cycle on $\mathbb{Z}_m^2$, that every arc is a valid Cayley digraph arc, and that the three cycles partition all $3m^3$ arcs.

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